

HIGH LEVEL SOLUTION INTENT

Commercial Credit Brain.

A reusable Accenture accelerator: one governed brain across the entire commercial credit lifecycle, built for a **regulated commercial bank** on **AWS**. Proven on a flagship lender, it becomes a repeatable asset for the next.

SUBJECT	ESTATE	STATUS
The intended target architecture, approach, components, and net-new prerequisites for a regulated commercial credit brain, before detailed design.	Amazon Bedrock and Bedrock AgentCore, Snowflake and Cortex, Bedrock Guardrails with Automated Reasoning. Roughly 30,000 users.	Stage 0 solution intent. Two MCP servers built and running; the full governed topology is the proposed build. Conditional-go, gates in Section 10.

01 Solution intent

EXECUTIVE SUMMARY

Accenture builds one brain across the entire commercial credit lifecycle, not a point tool for a single task. It is built for a regulated commercial bank from the first line. Proven on a flagship lender, the same brain becomes a reusable Accenture asset that every other lender can adopt.

The intent is deliberately broad. The brain spans the whole of commercial credit, from origination through spreading, rating, underwriting, approval, documentation, servicing, monitoring, and workout, with one governed orchestrator beneath every function rather than a separate tool bolted onto each. It is built for a regulated lender from the start: roughly **30,000 users**, per-user entitlements, a complete and queryable audit trail, model-risk governance on anything that affects a credit outcome, and policy checks on every decision the model touches.

The approach is to start narrow and grow agent by agent. The first agent, the **credit memo**, is already built and running. It earns its place by being useful on day one and by proving the substrate everything else reuses. Each subsequent agent, beginning with **Early Warning** monitoring, costs a fraction of the first, because it inherits the same data foundation, the same guardrails, and the same governed connections. One narrow win compounds into a full commercial credit brain.

That is what makes this an **asset, not a one-off**. The lifecycle, the ontology, the governance pattern, and the credit memo agent itself are not bank-specific accidents; they are the shape of commercial lending at any institution. The flagship lender is where the pattern is proven and hardened. After that, the work to stand it up at the next lender is configuration over a known asset, not a new build.

THE ASSET THESIS

This document describes the brain as it is intended for the first lender. The same brain, with the same lifecycle spine and governance, is the reusable Accenture banking asset. **Flagship first, then repeatable**. The bank gets a credit brain built to its estate; Accenture gets reusable commercial-credit IP it can take to the next regulated lender.

HOW TO READ THIS DOCUMENT

Section 02 sets the business context. Section 03 states the brain pattern and methodology. Section 04 is the target AWS architecture, the topology, the credit lifecycle, and one request end to end. Section 05 covers the two use cases. Section 06 is how the brain learns and governs itself, including model-risk governance. Section 07 is the solution components and the net-new prerequisites. Section 08 is the delivery approach and phasing with Stage 1 success criteria. Section 09 is the non-functional intent. Section 10 is assumptions, the open log, the data classification, the risk register, and what is out of scope. Section 11 is status and the next step. Section 12 is the team and the stakeholders whose sign-off the gates require.

02 Business context and drivers

THE MODEL IS RENTED; THE BRAIN IS OWNED

Frontier models are becoming a commodity. Every lender can rent the same capable model. The compounding differentiator is the institution's own credit knowledge: its lending policy, its precedent, and the lessons it captures on every decision. The model is rented. The brain is owned.

A capable model is necessary but not sufficient. Claude answers questions; it does not, on its own, hold a bank's covenant ontology, reach its systems of record under the right officer's entitlements, draft a credit memo against current lending policy, or leave an audit trail a regulator will accept. Those capabilities live in the layers assembled around the model. That assembly, inside a controlled trust boundary, is the brain, and it is the part a competitor cannot rent.

Why this matters now, for a commercial lender

Commercial credit decisions are slow, knowledge-intensive, and unevenly captured. A seasoned credit officer carries precedent in their head; when they retire, it leaves with them. Spreading, rating, and memo drafting consume senior time that could go to judgment. And every regulator expects the institution to show, for any decision, exactly what informed it. A brain addresses all three at once: it puts the institution's accumulated knowledge in front of every officer, it does the assembly work so people spend their time deciding, and it leaves a complete, auditable trail of why.

The opportunity, on published benchmarks

Independent research sizes the prize and frames it honestly as a benchmark, not a promise. McKinsey reports that generative AI can lift credit-analyst productivity on memo preparation and shorten decision cycles materially, and that a large share of knowledge work in the function is automatable. The point of a Stage 0 intent is to commit to **measuring the bank's own baseline**, cycle time per memo, senior hours per decision, consistency of policy application, and to hold the pilot to a target against that baseline rather than to a vendor's number.

THE DRIVERS, STATED PLAINLY

Speed and consistency in credit decisions. Capture of institutional knowledge so it compounds instead of walking out the door. A defensible audit trail on every decision a regulator can inspect. And a foundation that grows cheaper per use case as it spans the lifecycle. The bank that owns its **brain** compounds an advantage no rented **model** can level.

The strategic consequence follows directly. If the model is a commodity and the brain is the moat, then the institution should invest in the brain: the tools, the skills, the ontology, the governed data, and the learning loop that makes every decision sharpen the next. That is the work this solution intent describes.

03 Solution overview and methodology

READ DIRECT, ACT THROUGH THE BRAIN

The brain is a custom MCP server that holds the bank's knowledge, memory, skills, governance, and per-user access. Each officer connects to it through one connector. Claude Cowork is the harness. The doctrine is simple: read direct, act through the brain. Start narrow with the credit memo, proven, and grow agent by agent along the lifecycle.

The brain is not an abstraction. It is software you build and run: a custom MCP orchestrator that sits between Claude and every system of record beneath it. It holds the skills, the memory, the project context, the governance gates, and the per-user access, and exposes all of it to Claude through a single connector. The brain reaches each tool server as an MCP client, on the user's behalf, so one audited surface fronts every system of record.

The five-layer methodology

The pattern decomposes into five layers assembled around the model, the methodology the brain implements and the checklist any instantiation is reviewed against.

LAYER	WHAT IT IS, IN THIS BRAIN
1 · Agent ensemble	Task agents on one brain: credit memo, then Early Warning, then one per lifecycle stage. Switched on from a prioritised backlog as configuration, not forked builds.
2 · Specialized models	Claude via Amazon Bedrock for reasoning and drafting; Snowflake Cortex for in-database inference; room for custom scoring models where one fits.
3 · Domain ontologies	Borrower, facility, covenant, collateral, obligor, exposure. A Snowflake semantic layer today; a knowledge graph for entity resolution later.
4 · Secure data foundation	Snowflake as the governed store with Cortex Search for hybrid retrieval, in-boundary, no shadow vector index and no new copy of bank data.
5 · AI lifecycle management	Guardrails and Automated Reasoning on every call, versioning and evaluation, model-risk governance on anything that affects a credit outcome, observability, the learning loop.

THE DOCTRINE · READ DIRECT, ACT THROUGH THE BRAIN

Reads may federate directly through turnkey, per-user-delegated connectors: managed MCP servers reaching Snowflake and nCino under the officer's own entitlements. **Everything that acts goes through the brain:** orchestration, writes, agents, and memory. The read path inherits existing permissions and adds no new risk. The write path is where new risk enters, so the write path is the path that gets the brain, the audit, and the gate.

Why the brain hop wins for a regulated lender

Routing through the brain buys four things a direct-to-tool design cannot: one connector and one authentication per user; central audit on every tool call; per-user credential brokering, so the brain acts as the user, never a more-privileged deputy; and shared memory, context, and skills across every agent. The cost is one additional hop, negligible against inference, and the brain becomes critical infrastructure kept highly available.

04 Target architecture: the AWS estate

HIGH LEVEL TOPOLOGY

Officers work in Claude Cowork, which connects to the brain through one connector. The brain, in AgentCore Runtime, orchestrates the tool MCPs over Snowflake, with Claude served by Bedrock and governed by Guardrails. All inside the bank's own AWS account.

■ Build now ■ Stage 1 governed path ■ R1 / scale growth path ■ Future state

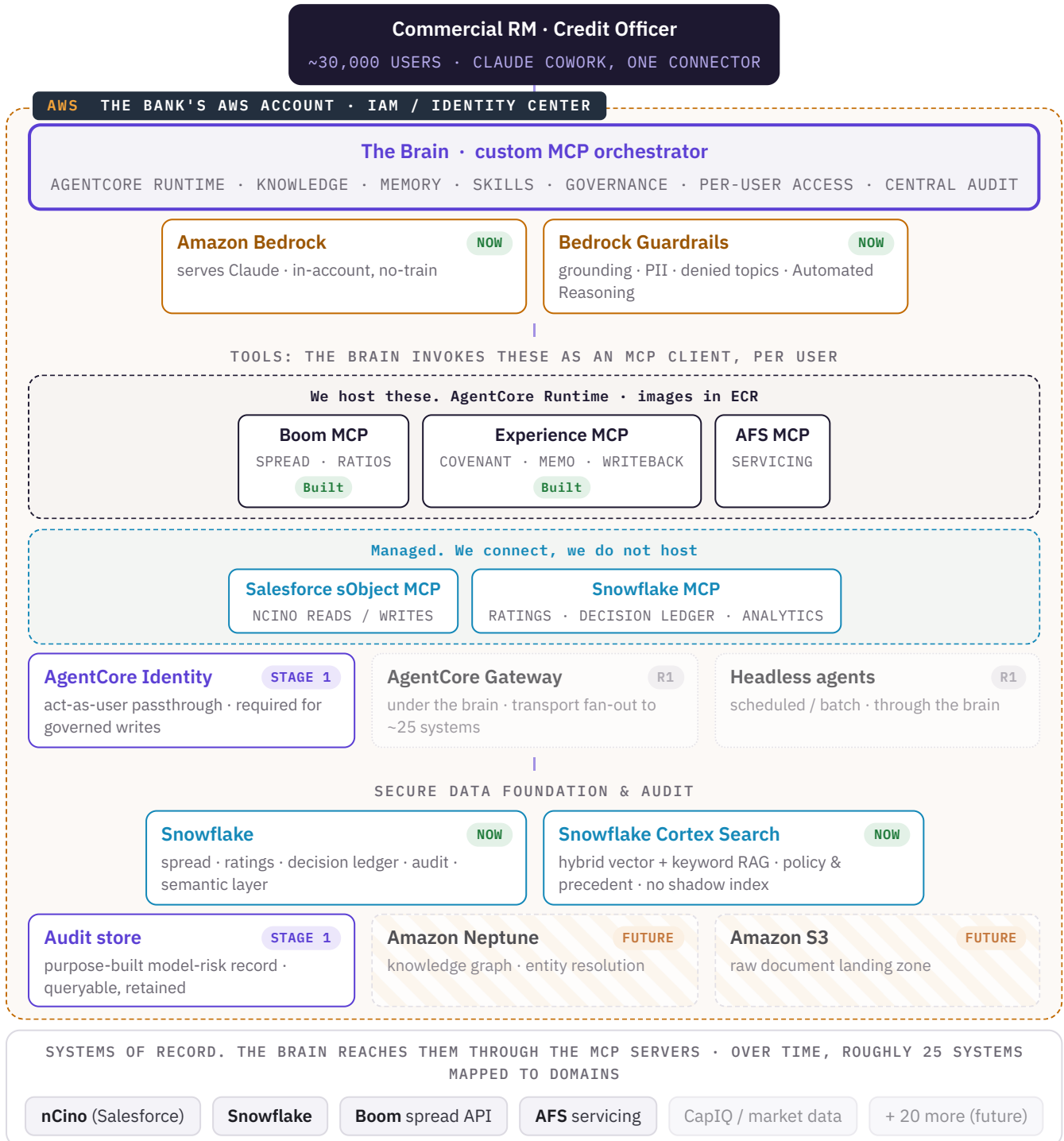


FIG. 1. THE TARGET SHAPE, INSIDE THE BANK'S OWN AWS ACCOUNT. THE BRAIN INVOKES EVERY TOOL MCP PER USER WITH CENTRAL AUDIT. TWO SERVERS ARE BUILT (BOOM, THE MEMO SERVER). AGENTCORE IDENTITY AND THE AUDIT STORE ARE STAGE 1; GATEWAY AND HEADLESS AGENTS ARE R1, BENEATH THE SAME BRAIN.

04 The credit lifecycle, one brain beneath it

SPINE · BUILT / NEXT / ROADMAP

The brain spans the whole commercial credit lifecycle. Each stage is a system of record today; each becomes an agent on the same brain over time. The honest state is marked at every stage: two stages are built, one is next, the rest are on the roadmap.

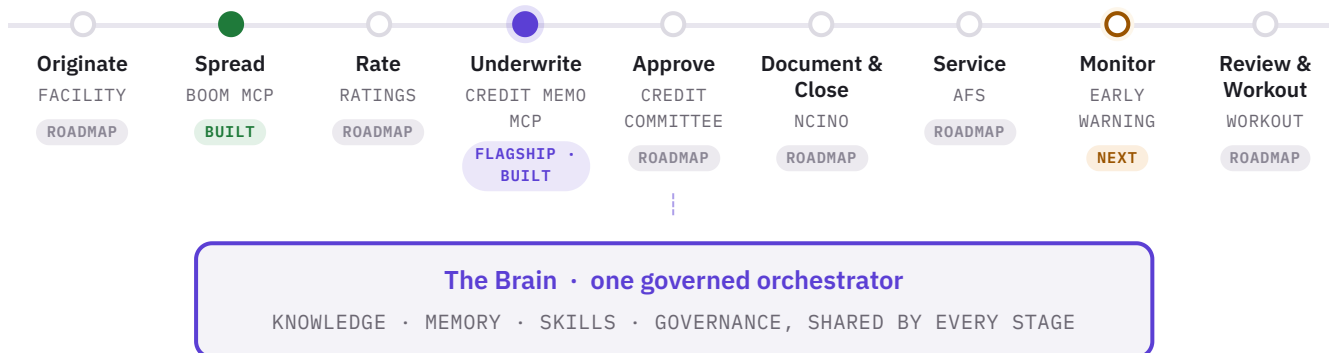


FIG. 2. THE COMMERCIAL CREDIT LIFECYCLE, WITH ONE BRAIN BENEATH EVERY STAGE. SPREAD (BOOM MCP) AND UNDERWRITE (THE CREDIT MEMO MCP, THE FLAGSHIP AGENT) ARE BUILT AND RUNNING. MONITOR (EARLY WARNING) IS THE NEXT AGENT. THE REMAINING STAGES ARE THE ROADMAP, EACH A SYSTEM OF RECORD TODAY THAT BECOMES AN AGENT ON THE SAME BRAIN OVER TIME.

What happens on one request

The topology resolves into one repeatable path. An officer asks for a credit memo in Cowork; the brain assembles it as the user, grounded against the institution's own policy and precedent, governed and audited end to end.

- 1 The officer asks in Cowork.** A request to draft or update a credit memo for a borrower. COWORK · MCP CLIENT
- 2 Cowork connects to the brain** through one connector. The brain authenticates the officer and adopts their access set for the whole request, acting as the user through AgentCore Identity. IAM · AGENTCORE IDENTITY
- 3 The brain retrieves context** in-boundary: borrower history, lending policy, and lessons from prior memos, via Snowflake Cortex Search, no copy leaving the account. SNOWFLAKE · CORTEX
- 4 The brain calls the tools as the user:** Boom for spreads and ratios, the memo server for covenant grade and section drafting, Snowflake and nCino for facility and ledger data. BOOM · MEMO · ORIGINATION
- 5 Bedrock serves Claude; Guardrails govern the call:** contextual grounding, PII, denied topics, and Automated Reasoning against encoded lending policy. BEDROCK GUARDRAILS
- 6 Every step is written to the audit store,** and the officer reviews and decides. The draft routes through human approval; nothing is final until a credit officer signs off. AUDIT · HUMAN-IN-THE-LOOP

05 Use cases

CREDIT MEMO · EARLY WARNING

Two agents on one brain. The first builds the foundation with a human in the loop; the second runs on top of it, headless, for a fraction of the cost. Both share the same brain substrate.

UC1 · MVP · BUILT

Credit Memo

A long-running agent assembles a commercial credit memo. It pulls borrower financials and spreads, facility and covenant data, and supporting context from Snowflake, drafts the sections of the memo, and routes them through human review and approval.

The brain wraps each call with context retrieved at drafting time, the borrower's history, current lending policy, and lessons captured from prior memos, so the draft is grounded in what the institution already knows. The output routes through human approval before anything is final. Every memo makes the next one better: lessons captured on this engagement promote into the memory the next draft retrieves from.

Controls.

- Contextual grounding so claims trace to source data
- PII filtering on every model call
- Automated Reasoning against encoded lending policy
- Human approval gates before anything is final

The agent drafts; a credit officer decides. The output is a starting point that is faster to review, never an unreviewed decision.

UC2 · R1 · NEXT

Early Warning

Monitoring agents watch portfolio signals: covenant breaches, financial deterioration, market and news events. They run scheduled and event-driven, headless, with no user at the keyboard, and surface risk indicators and alerts to the relevant officers.

Early Warning runs on AgentCore Runtime as scheduled and headless agents on the same brain. It reuses the data foundation, the retrieval, and the guardrails the credit memo built, so the agent is mostly new monitoring behaviour over an existing substrate.

Controls.

- Scheduled and event-driven headless runs
- Full auditability of why an alert fired
- The same grounding and policy checks as UC1
- Alerts routed, not actioned automatically

Every alert carries its reasoning trail, so a reviewer can see exactly which signal triggered it and why.

SHARED SUBSTRATE

UC2 is cheaper because UC1 built the foundation. The Snowflake data foundation, the Cortex retrieval, the guardrails, the governed connections to nCino and the lending systems, and the audit trail are all in place once the credit memo agent ships. Early Warning adds monitoring logic and scheduling on top. The second use case is mostly new agent behaviour over an existing brain, which is the entire economic argument for the asset.

06 How it learns and governs itself

MEMORY · LEARNING · MODEL RISK

The brain compounds. Every memo captures what worked; reuse is the signal that a lesson is worth keeping; a governed gate graduates proven lessons into versioned skills and precedent. Memory lives in the Snowflake decision ledger with provenance and decay. This loop is the moat, and it is governed as model risk.

Memory, learning, and the gate

Memory

Decisions, lessons, and provenance are written to the **decision ledger in Snowflake**, scoped per borrower, portfolio, and policy. Every item carries provenance, so any retrieved fact traces to where it came from and why it is trusted. Unused, unpromoted memory **decays**: it is archived, never silently deleted.

Learning

The brain captures **what worked on each memo** at the lowest true scope. Reuse is the signal: every retrieval that gets used bumps the source's reuse count, which drives ranking and protects proven knowledge against decay. What the institution actually relies on rises; what nobody uses settles out.

Skill promotion

A captured lesson is **proposed**, then passes a **governance gate**, a human reviewer plus a Bedrock Guardrails and Automated Reasoning policy check, and only then **graduates** into a versioned skill or precedent every future memo starts with. Promotion is governed, never automatic.

PROMOTION IS A MODEL-RISK EVENT · SR 11-7

Any promoted skill, prompt, or rule that can change a credit outcome is a **model under model-risk management** (SR 11-7, OCC 2011-12), not a content edit. Each candidate gets a **model inventory entry**, an **independent validation** before promotion, a documented **approval by model risk**, and **ongoing monitoring** after it ships. The promotion gate is the technical enforcement point of that governance, not a substitute for it.

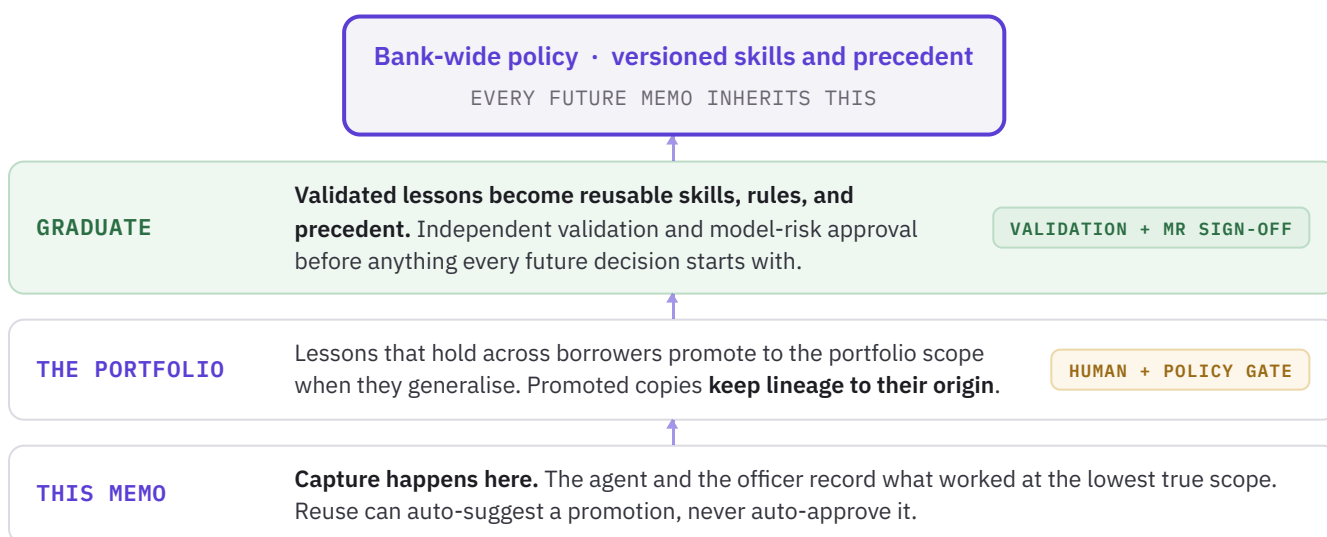


FIG. 3. THE PROMOTION LADDER. LESSONS GRADUATE UPWARD ONLY THROUGH A GOVERNED GATE AT EACH RUNG; THE TOP GATE IS INDEPENDENT VALIDATION PLUS A MODEL-RISK SIGN-OFF. PROMOTED COPIES KEEP LINEAGE TO THEIR ORIGIN.

07 Components and net-new prerequisites

WHAT IT IS MADE OF · WHAT MUST BE PROCURED

The solution decomposes into seven component groups. Most run on platforms the bank already operates. This section lists the components, then itemizes strictly the net-new licenses and enablements the brain requires beyond the assumed baseline estate.

Solution components

COMPONENT	ROLE	BUILD OR BUY
The brain custom MCP orchestrator	Holds knowledge, memory, skills, governance, per-user access; brokers every tool call as the user with central audit.	BUILD & HOST
Hosted tool MCPs Boom, Memo, Servicing	Spread and ratios, covenant grade and memo writeback, servicing. Two of three built and running.	BUILD & HOST
Managed tool MCPs Salesforce, Snowflake	nCino reads and writes; ratings, decision ledger, analytics. Connect, do not host.	CONNECT
Model substrate Bedrock + Guardrails	Claude in-account with no training; Guardrails and Automated Reasoning on every call.	CONFIGURE
Data foundation Snowflake + Cortex	Governed store and in-boundary hybrid retrieval. No new copy, no shadow index.	CONFIGURE
Governance plane identity, audit, model risk	AgentCore Identity for act-as-user; purpose-built audit store; model inventory and validation.	STAND UP
Client channel Claude Cowork	The desktop harness officers work in; the MCP client that reaches the brain.	LICENSE

ASSUMED BASELINE ESTATE (ALREADY IN PLACE)

AWS account and sanctioned landing zone; IAM and Identity Center; Snowflake holding the bank's lending data; the core lending systems of record (loan origination on Salesforce, the ratings engine, the Boom spread API, the servicing platform). The list below is what is **net-new on top of that baseline**. Anything the bank already runs (for example Bedrock or Guardrails) drops off the procurement list and stays only as an assumption to confirm in Section 10.

07 Net-new prerequisites and licenses

STRICTLY NET-NEW, ONE LINE EACH

Each line is a distinct thing to procure or enable, with its type. **Enable** is a platform capability to switch on or a model access request and EULA. **Consumption** is metered pay-as-you-go usage. **License** is a paid license, edition uplift, or seat purchase. **Contract** is a commercial or legal agreement.

PREREQUISITE	TYPE	WHY IT IS NEEDED
MODEL AND GOVERNANCE		
Anthropic Claude on Bedrock per model, e.g. Opus / Sonnet	ENABLE CONSUMPTION	Model access request and EULA acceptance per model; in-account, no-train inference, metered per token.
Bedrock Guardrails	ENABLE CONSUMPTION	Grounding, PII, denied topics on every call. Drops off if the bank already runs it.
Bedrock Automated Reasoning Checks	ENABLE CONSUMPTION	The differentiating control: formal policy checks. Priced separately from standard Guardrails; the policy authoring is a workstream, not a config step.
Bedrock Model Invocation Logging	ENABLE CONSUMPTION	Captures the raw model-call record that feeds the purpose-built audit store.
RUNTIME AND ORCHESTRATION · BEDROCK AGENTCORE		
AgentCore Runtime	ENABLE CONSUMPTION	Hosts the brain and the custom MCP servers, session-isolated and long-running. Confirm regional GA.
AgentCore Identity Stage 1, not later	ENABLE CONSUMPTION	Act-as-user passthrough. Required before any write goes through the brain, to close the confused-deputy gap.
AgentCore Gateway R1 / scale	ENABLE CONSUMPTION	Transport fan-out and curation across many tool servers as the system scales to roughly 25 systems.
AgentCore Observability	ENABLE CONSUMPTION	CloudWatch and OpenTelemetry traces for every run, input to reasoning to output.
Amazon ECR	ENABLE CONSUMPTION	Container registry for the hosted MCP server images that AgentCore Runtime runs.

07 Net-new prerequisites, continued

DATA · CHANNEL · NETWORK

The remaining net-new items: the data-foundation enablements, the client channel and the agreements behind it, and the network and future-state additions.

PREREQUISITE	TYPE	WHY IT IS NEEDED
DATA FOUNDATION		
Snowflake Cortex Cortex Search + functions	ENABLE CONSUMPTION	In-boundary hybrid retrieval and in-database inference. Consumes credits. Confirm regional GA before committing the no-shadow-index design.
Snowflake edition uplift if currently Standard	LICENSE	Cortex and the governance features assume Enterprise or Business Critical as the baseline edition.
Purpose-built audit store model-risk record	CONSUMPTION	A queryable, retained store of every tool call, parameter, return, reasoning trace, model version, Guardrails result, and human override. A Snowflake schema or OpenSearch domain, not raw Bedrock logs.
CLIENT CHANNEL AND AGREEMENTS		
Claude Cowork enterprise seats ~30,000 users	LICENSE	The desktop harness. An Anthropic enterprise commercial track, priced and contracted separately from Bedrock inference.
Anthropic enterprise agreement & DPA	CONTRACT	The commercial and data-processing vehicle covering Cowork and any Anthropic-hosted interactive path.
NETWORK AND FUTURE STATE		
AWS PrivateLink endpoints R1 hardening	CONSUMPTION	Private transport between services. Hardening, not a prerequisite to start.
Amazon Neptune future	CONSUMPTION	Knowledge graph for cross-system entity resolution across borrower, guarantor, and exposure.
Amazon S3 landing zone future	CONSUMPTION	Raw document landing for ingestion ahead of structuring.

Pricing is consumption-led and not committed here; a Stage 1 cost model sizes inference, AgentCore, and Cortex usage against the pilot's volume. Items already running in the bank's estate drop from this list and move to the assumptions to confirm in Section 10.

08 Delivery approach and phasing STAGE 0 TO STAGE 1, THEN R1, THEN SCALE

One architecture across time. The brain is the constant; each phase adds reach beneath it, never a competing design. Stage 1 is the governed pilot, and it carries the controls a regulated lender requires from the first write, not later.

STAGE 1

Governed pilot

- Build the eval harness: golden test cases first
- Re-point the model to Bedrock, in-account
- Re-host the two built servers into AgentCore Runtime
- Stand the brain up in front, one connector per user
- AgentCore Identity for act-as-user on writes
- Author and validate the Automated Reasoning policy
- Stand up the purpose-built audit store; multi-AZ brain

R1

Scale the plumbing

- AgentCore Gateway under the brain for many systems
- Headless Early Warning monitoring agents
- PrivateLink as transport hardening
- Same brain, more reach beneath it

SCALE

Across the estate

- Roughly 25 systems of record mapped to domains
- New capability is configuration over the registry
- Neptune and S3 for the graph and landing zone
- Replicate the asset to the next lender

FIG. 4. THE PHASING. STAGE 1 ROUTES THE PROVEN CREDIT MEMO THROUGH THE BRAIN ON BEDROCK AND AGENTCORE WITH NO REWRITES, AND CARRIES IDENTITY, POLICY, AUDIT, AND HIGH AVAILABILITY FROM THE FIRST WRITE. R1 ADDS THE GATEWAY AND HEADLESS EARLY WARNING. SCALE EXTENDS ACROSS ROUGHLY 25 SYSTEMS, WHERE NEW CAPABILITY IS CONFIGURATION OVER THE REGISTRY, NOT A NEW BUILD.

STAGE 1 SUCCESS CRITERIA • MEASURABLE, AGAINST THE BANK'S OWN BASELINE

Credit-memo cycle time reduced against the measured baseline by an agreed target. A defined share of pilot memos drafted through the brain. Every drafted claim grounded and traceable to source. Zero cross-entitlement retrieval in an independent penetration test. One hundred percent of tool calls captured in the audit store with a reasoning trace. The Automated Reasoning policy passing its validation set. The human-approval gate enforced on one hundred percent of finalizations. Brain availability meeting its multi-AZ target through the pilot.

WHY THE ASSET REPLICATES

Because expansion is configuration over an agent registry, standing the brain up at the next lender is not a fresh build. The lifecycle spine, the ontology, the governance pattern, and the credit memo agent are reused; what changes is the bank's data foundation, entitlements, and policy. The flagship lender hardens the asset. Every lender after inherits it.

08 Stage 1 hardening: the path to green

WHAT TURNS CONDITIONAL INTO READY

An independent architecture review rates the corrected design amber: strong enough to enter Stage 1, not green until five controls are in place. Each is named here with an owner. The evaluation harness comes first; the rest is plumbing around a validated core.

CONTROL	WHAT IT ADDS	OWNER
Evaluation harness do this first	20 to 30 golden test cases: fixed borrower inputs, expected tool-call sequence, expected rationale, and expected Automated Reasoning triggers. The regression baseline every future promotion must pass, and the proof the policies actually fire.	Model Risk / Delivery
Tool-output integrity	Validate and sanitise MCP tool responses (Snowflake, nCino) before the model acts on them. Guardrails govern model input and output, not tool returns; a poisoned result that redirects the next tool call is otherwise uncontrolled.	CISO / Platform
Decision explainability	Surface the factor-level rationale: which data fields triggered which policy, producing which outcome. Exam-ready and fair-lending-ready, not "the model decided".	Model Risk / Credit Policy
Duty separation + rollback	A four-party promotion chain (build, review, independent validation, model-risk push) so a builder cannot self-promote, plus one-step rollback of any live skill within minutes.	Model Risk / Platform
Cost and continuity	Per-session and per-workflow token ceilings against runaway recursion, and a documented degraded-mode fallback for when the brain, a managed MCP, or the model backend is unavailable.	Platform / SRE

WHY THE HARNESS COMES FIRST

It validates whether the Automated Reasoning policies fire correctly, gives model risk a concrete artefact to sign, sets the regression baseline for every promotion, and surfaces the tool-output exposure above before it reaches production. The other four controls are built around it. This is the move that takes the design from amber to green.

09 Non-functional intent

IDENTITY · DATA · GOVERNANCE · RESILIENCE

The qualities the brain must hold to run inside a regulated lender, stated at the intent level. Identity per user, least privilege, no new copy of data, governance on every model call, a purpose-built audit record, high availability, and scale to roughly 30,000 users.

CONCERN	INTENT
Identity per user, act-as-user	AWS IAM and Identity Center for authentication. The brain adopts the caller's access set for the whole request and acts as the user through AgentCore Identity from Stage 1, so the brain is never a more-privileged deputy on any write.
Least privilege	Every tool call runs under the officer's own entitlements. The brain holds no standing super-privilege over the systems of record; it brokers, per user, with central audit on every call.
Data residency no new copy	No new store and no shadow vector index. Retrieval is in-boundary over Snowflake with Cortex Search. Bedrock inference runs in the bank's account with no training on its data. Every element sits inside the account boundary. Confirm Cortex regional GA before committing this.
Model channel Bedrock, in-account	Claude is served by Amazon Bedrock inside the bank's account , not by each user's personal Claude seat. The harness is Cowork; its inference backend is the bank's Bedrock. This is the one change from the prototype, and it keeps bank data out of externally hosted inference.
Governance the differentiating control	Bedrock Guardrails on every model call: contextual grounding, PII filtering, denied topics, and Automated Reasoning against encoded lending policy, a formal, checkable yes or no against the rules.
Auditability model-risk grade	A purpose-built, queryable audit store records every tool call, parameter, return value, reasoning trace, model version, Guardrails result, and human override. Raw Bedrock logs feed it; they are not the record themselves. Any decision can be reconstructed for a regulator.
Resilience critical infrastructure	At 30,000 users the brain is critical infrastructure. Multi-AZ deployment, a circuit breaker on downstream tool failures, and a documented disaster-recovery posture, so brain unavailability is a managed event, not a business-continuity incident. Per-session token ceilings cap runaway cost, and a degraded-mode fallback keeps underwriters working when a dependency is down.
Scalability	Roughly 30,000 users. AgentCore Runtime is session-isolated and long-running; agents are switched on progressively from a prioritised backlog as configuration over the registry.
Transport	Streamable HTTP over HTTPS, with SigV4 or bearer auth. No tunnel. AWS PrivateLink is hardening of the transport, added at R1.

THE LINE A SECURITY REVIEWER CARES ABOUT

Retrieval is permission-correct by construction. Every query carries the caller's access set, and the scope filter runs **before** candidates are selected from the index, not as a post-filter. A caller can never see an item outside their access set, because items outside it are never candidates. The write path, where new risk enters, is the path that gets the brain, act-as-user identity, the audit, and the gate.

10 Assumptions, gates, and risks

WHAT WE ASSUME · WHAT MUST BE CONFIRMED

A Stage 0 intent is only credible if it is honest about what it assumes and what it must guard against. The items below are owned by the institution's security, platform, data, and model-risk teams, and should be settled before detailed design.

Assumptions and dependencies

Bedrock in-account	Claude is served by Amazon Bedrock inside the bank's AWS account, with no training on the bank's data . This is the model-channel assumption the whole architecture rests on.
No new copy of data	The bank accepts no new store and no shadow vector index as the storage model. Retrieval stays in-boundary over Snowflake and Cortex.
Systems of record reachable	nCino, Snowflake, Boom, and AFS are reachable over governed MCP transport under per-user entitlements. The ~25-system map is a future dependency, not a day-one one.
Landing zone	Compute for the brain is provisioned through the bank's sanctioned landing-zone process, with an approval trail, rather than a self-managed box.

Open items to confirm before Stage 1 (the gates)

AgentCore availability	Confirm AgentCore Runtime, Identity, and Gateway GA status, region, and pricing in the bank's region. The Stage 1 design depends on Identity being available.
Cortex regional GA	Confirm Cortex Search is GA in the bank's Snowflake region. The no-shadow-index architecture depends on it; if not, a fallback retrieval design is needed.
Snowflake edition	Confirm the current edition. Cortex and the governance features assume Enterprise or Business Critical.
Guardrails and AR	Confirm Bedrock Guardrails and Automated Reasoning are enabled, and scope the policy-authoring workstream to encode lending policy.
Model backend path	Confirm the Cowork harness uses the bank's in-account Bedrock as its inference backend (rather than the agent running server-side in AgentCore). The no-bank-data-to-external-inference guarantee depends on this.
Evaluation harness	The golden-test-case harness exists and passes before Stage 1 go-live. It is the artefact model risk signs, and the baseline every promotion must clear.

10 Data classification and risk register

OWNED, NAMED, MITIGATED

Data classification (Stage 0 stub, to be completed in design)

DATA ELEMENT	CLASSIFICATION	HANDLING INTENT
Borrower financials and spreads	Confidential / NPI	In-account, no copy. Retrieved in-boundary; never leaves the account.
Facility, covenant, collateral data	Confidential	In-boundary over Snowflake under per-user entitlements.
Lending policy and precedent	Internal / Confidential	Cortex Search in-boundary; the grounding source, not training data.
Decision ledger and memos	Confidential, audit-retained	Snowflake ledger with provenance; retained per records policy.
Personal information (PII)	Restricted NPI	Guardrails PII filter on every model call; masked in logs.
Model prompts and responses	Confidential	Logged to the audit store; no training on bank data.

Risk register

RISK	CONTROL	OWNER
Ungrounded output	Contextual grounding and Automated Reasoning against lending policy, plus a human approval gate before anything is final.	Model Risk / Credit Policy
Confused deputy	Per-user credential brokering and AgentCore Identity act-as-user from Stage 1, with central audit on every call.	CISO / IAM
Skill promotion as model risk	Model inventory entry, independent validation, and model-risk sign-off before any skill that affects an outcome is promoted (SR 11-7).	Model Risk / Validation
Poisoned or wrong memory	The curation gate: propose, then a human or policy promotes; provenance and lineage on every item; decay on unused memory.	Credit Policy
Silent regression	Versioning of agents, skills, and prompts, with evaluation gates a change must pass before it ships.	Delivery / MLOps
Brain availability	Multi-AZ deployment, circuit breaker on downstream failures, documented DR. Brain treated as critical infrastructure.	Platform / SRE
Data residency	No new copy, in-boundary retrieval, in-account inference; Cortex regional GA confirmed before commit.	Data Governance

OUT OF SCOPE FOR THIS INTENT

No autonomous credit decisions; the agent drafts, a credit officer decides. No new system of record and no replacement of core banking. No customer-facing channel. No fine-tuning or training on the bank's data. No data egress outside the bank's AWS account. The ~25-system map and the future graph and landing zone are roadmap, not Stage 1.

11 Status and next step

HONEST · WHAT IS BUILT, WHAT IS PROPOSED

Stated plainly, with no overclaim. The feasibility verdict is conditional-go: the architecture is sound; the conditions are the Stage 1 controls in Section 08 and the gates in Section 10. Two MCP servers are built and running. Routing them through the brain on the full governed topology is the proposed build.

CONDITIONAL GO

The approach is feasible and defensible for a regulated commercial credit brain. The conditions to clear before a Stage 1 build: **act-as-user identity on every write**, the **Automated Reasoning policy authored and validated**, the **purpose-built audit store**, **multi-AZ resilience**, and the **model-risk inventory and validation** for anything that affects a credit outcome, plus the **five Stage 1 hardening controls** in Section 08 (the evaluation harness first). None is a blocker; each is a named piece of Stage 1.

BUILT AND RUNNING

The Credit Memo MCP (the flagship Underwrite agent) and **the Boom MCP** (Spread, spreads and ratios). Both run today. The working demo has Cowork calling the credit-memo server directly, with no brain in between, the honest stepping stone.

PROPOSED BUILD

The full governed topology: the brain in AgentCore Runtime in front of the tool servers, served by Bedrock, grounded by Cortex, governed by Guardrails, with act-as-user identity and a model-risk audit store, one connector per user.

NEXT AGENT

Early Warning: headless, scheduled portfolio monitoring on the same brain. It reuses the foundation the credit memo built, so it is mostly new monitoring behaviour over an existing substrate.

TODAY'S DEMO STATE, STATED HONESTLY

The working demo has Cowork calling the credit-memo MCP directly, with no brain in between. That is the honest stepping stone, not the target. The target is the brain in the middle: one connector per user, the brain brokering every downstream call as an MCP client, and a single audited surface over all of the tool servers. Nothing in this document claims the full topology is live. Two servers are built; the topology is the proposed build.

THE NEXT STEP

A **scoped Stage 1 pilot:** route the credit memo slice through the brain on the bank's estate. Re-point the model to Bedrock, re-host the credit-memo and Boom servers into AgentCore Runtime, stand the brain up in front with act-as-user identity and the audit store, and author the Automated Reasoning policy. This is the smallest move that proves the target shape end to end, and it reuses what is already built. Prove it on the memo; grow it along the lifecycle; replicate it as an Accenture asset for the next lender.

THE INTENT, IN ONE LINE

One **brain** across the whole commercial credit lifecycle, hosted in **AgentCore Runtime**, reached by Cowork through one connector, brokering per-user calls to MCP servers over **nCino**, **Boom**, **AFS**, and **Snowflake**, served by **Bedrock**, grounded by **Cortex**, governed by **Bedrock Guardrails and Automated Reasoning**, audited to model-risk grade, all inside the bank's AWS account. **No new copy of client data.**

12 Team and stakeholders

DELIVERY • SIGN-OFF

The people behind this solution intent, and the bank-side stakeholders whose alignment the Stage 1 gates require.

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Stakeholders whose sign-off the gates require

ROLE	WHAT THEY OWN IN THIS INTENT
CISO / Security	Identity, act-as-user, least privilege, the trust boundary, and the penetration-test criterion.
CRO / Model Risk lead	SR 11-7 model inventory, independent validation, and approval of anything that affects a credit outcome.
Credit Policy owner	The lending policy encoded into Automated Reasoning, and the curation gate on promoted precedent.
Data Governance owner	Classification, residency, the no-new-copy storage model, and Cortex regional confirmation.
Cloud Platform / Landing Zone	AgentCore enablement, the account boundary, multi-AZ resilience, and the provisioning trail.

TAKE IT FURTHER

This is a Stage 0 solution intent, prepared for discussion. The natural next move is a scoped Stage 1 pilot on the bank's estate: the credit memo slice through the brain, one stack, one slice, grown along the lifecycle. The same brain, hardened on the first lender, becomes the reusable Accenture asset for the next regulated lender.

Prepared as a high level solution intent (Stage 0). It states, at a high level, what is intended to be built, why, the target architecture and approach, the components, and the net-new prerequisites, before detailed design. The assumptions and gates in Section 10, particularly Bedrock in-account, the no-new-copy storage model, AgentCore and Cortex regional availability, and the model-risk governance of promoted skills, are owned by the institution's security, platform, data, and risk teams, and should be settled before detailed design. Two MCP servers are built and running; the full topology is the proposed build. This document describes a pattern and its intended instantiation; it is not a commitment to a specific implementation.